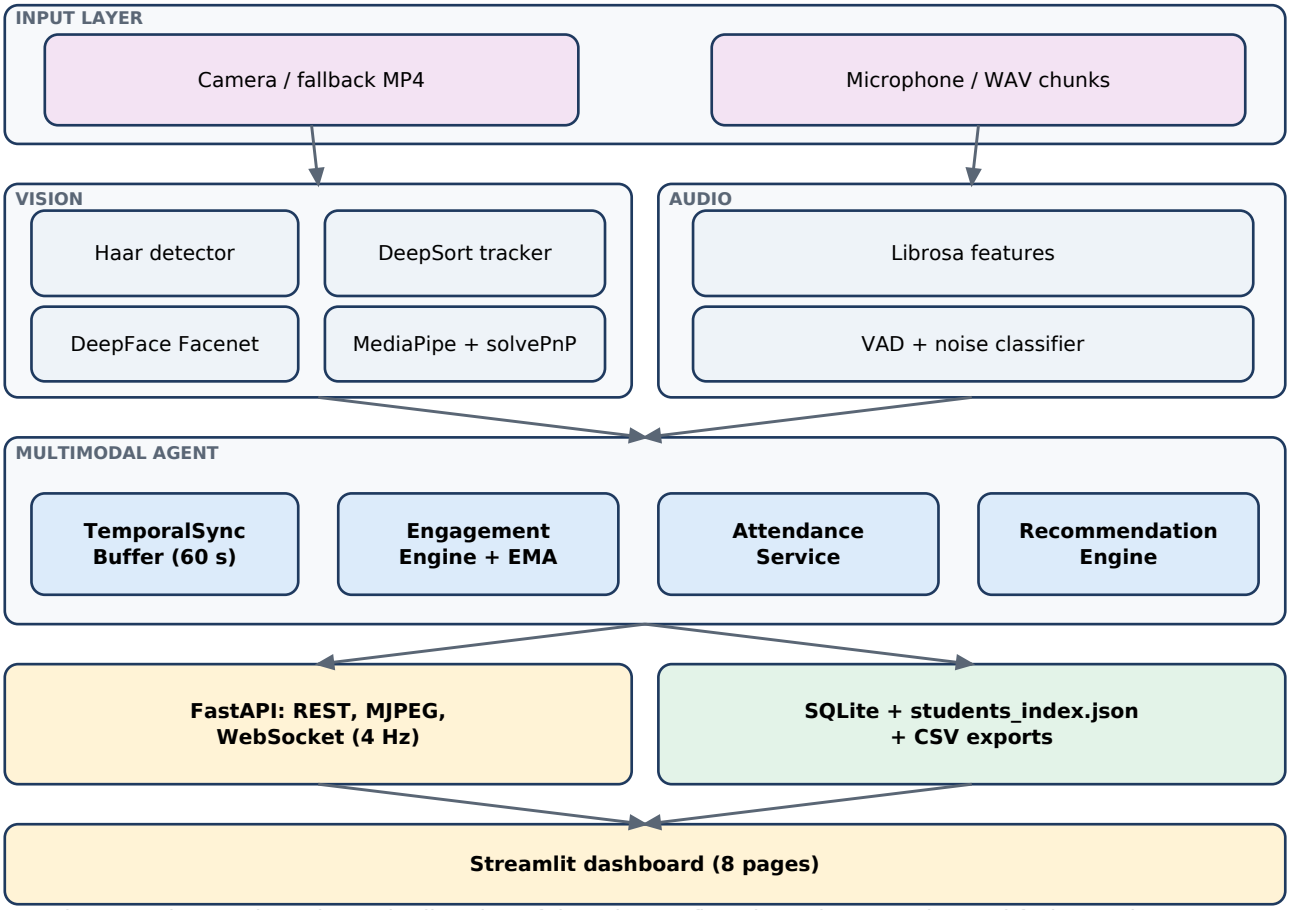
Solution Architecture

Multimodal Attendance & Engagement Agent • Nirbhay Dubey | B.Tech CSE (Year 1)

The architecture is decoupled: heavy vision and audio processing live in a FastAPI backend, and a Streamlit dashboard polls the backend. This prevents frame drops and UI freezes that occur when real-time vision runs inside Streamlit.

1. Layered Architecture



Arrows between layers show the main direction of data: inputs flow through perception and fusion to the API, storage, and dashboard.

2. Layer Responsibilities

Layer	Modules	Responsibility
Input	services/camera_service.py	Capture frames at 640x480 with MP4 fallback; toggle camera on demand
Vision	services/face_detection.py, face_recognition.py, head_pose_gaze.py	Detect, track, recognize, estimate pose and gaze
Audio	services/audio_analysis.py	Decode 16-bit PCM, extract features, classify acoustic state
Fusion and reasoning	multimodal_fusion.py, engagement_engine.py, attendance_service.py, reasoning_engine.py	Align signals, score engagement, qualify attendance, generate tips
Backend	api/main.py	REST endpoints, MJPEG video, WebSocket telemetry, session control
Persistence	database/models.py, crud.py, database.py	SQLAlchemy models for 5 tables on SQLite
Frontend	dashboard/app.py, dashboard/views/*	Navigation and 8 modular pages

3. Component Communication

From	To	Method	Data
CameraService	Face detector	Function call	BGR frame
Face tracker	Face recognizer	Function call	Track ID + face crop
AudioAnalyzer	Sync buffer	REST POST /api/audio/chunk	State, RMS, dBFS, SNR
Sync buffer	Engagement + Attendance	In-memory object	Aligned observation per student
Engagement + Attendance	Recommendation engine	Function call	Class averages, gaze ratio, attendance rate
FastAPI	Streamlit	HTTP REST, MJPEG, WebSocket	JSON status, video, telemetry
FastAPI	database/crud.py	SQLAlchemy session	Sessions, attendance, recommendations

4. Database Design (5 Tables)

Table	Key fields	Purpose
students	student_id (unique), name, roll_no, photo_path, consent_given	Enrolled identities and consent flag
sessions	session_id (unique), title, course_code, start/end time, is_active	Lecture sessions
attendance_records	session_id, student_id, track_id, time_present_sec, presence_ratio, status, mean_engagement	Final attendance per student per session
person_metrics	yaw, pitch, roll, ear, gaze_score, engagement_score, tier, audio_state	Per-frame telemetry schema (not written during live streaming)
recommendations	session_id, trigger_type, suggestion_text, priority	Triggered teaching suggestions

5. Design Decisions

- **Decoupled UI and backend:** keeps vision at full speed and the UI responsive.
- **In-memory buffers during a session:** avoids SQLite lock contention at 30 fps; only session summaries are committed.
- **Graceful fallbacks:** synthetic video when no camera, QUIET audio when no microphone chunks, bounding-box pose when the landmark model is missing.
- **Privacy by design:** local-only processing, consent flag enforced in code, no raw video stored.